Atoms, isotopes & models

Count particles

1 Nucleus & charge

Protons +1 and neutrons 0 each have relative mass 1. Electrons −1 have tiny mass.

Atomic number = protons. Neutrons = mass number − atomic number.

Na-23: 11p, 12n, 11e. Na^+: 11p, 12n, 10e.

2 Size & isotope averages

Atom radius ≈ 10^{-10} m; nucleus ≈ 10^{-14} m or less. Almost all mass is nuclear. **1 nm = 10^{-9} m.**

Isotopes: same protons, different neutrons. A_r weights each isotope by abundance.

75% mass 35 + 25% mass 37 → **35.5.**

Arrange electrons

3 Shells & position

Lowest shells fill first: **2,8,8,2** for first 20 elements. O: 2,6; Na: 2,8,1; Ca: 2,8,8,2.

Period = occupied shells. Same main group = same outer electrons → similar reactions.

Metals generally lose electrons; non-metals gain/share.

4 Atoms → ions

Losing negative electrons produces a **positive ion;** gaining them produces a negative ion.

The number of protons stays unchanged: it remains the same element.

Full outer shells help explain the stability of noble gases.

Follow the evidence

5 Scattering conclusions

Most alpha particles pass → atom mostly empty space.

Some deflect → nucleus is positive and repels positive alpha particles.

Very few return → tiny, dense nucleus holds most mass; contradicts spread-out positive charge.

6 Changing models

Indivisible spheres → discovery of electron → plum pudding. Scattering → nuclear model.

Bohr: specific electron shells. Later work: protons. Chadwick: neutrons.

New experimental evidence can replace a scientific model.